

CHAPTER 08 Organic compounds

At the end of this chapter, you will appreciate the significance of the compounds that are formed between carbon and hydrogen. Carbon is only the eleventh most abundant element in the universe, yet it forms more compounds than all other elements except hydrogen. Carbon is present in living things and many non-living things.

The prevalence of carbon in living things is why the study of the compounds of carbon and hydrogen is known as organic chemistry. You make use of these compounds in so many aspects of your life: from the polymer case of your telephone to the pasta that you may eat for dinner tonight.

In this chapter, you will start to explore some of the many families of carbon compounds. However, you will only scratch the surface of a varied and fascinating area of chemistry.

Science understanding

- hydrocarbons, including alkanes, alkenes and benzene, have different chemical properties that are determined by the nature of the bonding within the molecules
- IUPAC nomenclature is used to name straight and simple branched alkanes and alkenes from C_1 – C_8
- molecular structural formulae (condensed or showing bonds) can be used to show the arrangement of atoms and bonding in covalent molecular substances
- alkanes, alkenes and benzene undergo characteristic reactions such as combustion, addition reactions for alkenes and substitution reactions for alkanes and benzene

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8.1 Alkanes

Carbon forms more compounds than all other elements combined. This is because:

- carbon has four valence electrons so carbon can potentially form covalent bonds with four different atoms
- carbon atoms can form strong covalent bonds with other carbon atoms
- the covalent bonds formed can be a combination of single, double or triple bonds.

Because of this, carbon can bond to itself to form molecules of varied length and shape. The different structures have different properties and applications.

Carbon-based molecules are all around you. Caffeine, petrol, pesticides, plastics and artificial flavours are all carbon-based compounds. Many of these compounds are produced from **crude oil**. Crude oil is produced by the effects of heat and pressure on dead animals, plants and microorganisms trapped in the Earth's crust, buried beneath **sediment** formed over millions of years. The branch of chemistry that studies the chemistry of carbon compounds is known as organic chemistry. This section focuses on crude oil as a source of carbon compounds and on compounds found in crude oil that are formed between carbon and hydrogen: the **hydrocarbons**.

CRUDE OIL

The amount of carbon on Earth is essentially fixed, with most of the carbon atoms on Earth having been here for billions of years. It is the location of these carbon atoms that changes over time. Whether they are currently part of a sunflower seed, a molecule of carbon dioxide, a human being or a plastic chair, you can be certain that the carbon atoms have been in a different compound previously.

Origin of crude oil

Fossil fuels, such as coal, oil and natural gas, come from the remains of plants and animals. When prehistoric marine microorganisms, such as bacteria and plankton, died and were buried by sands millions of years ago, these organisms accumulated as organic sediment and gradually became part of the Earth's crust. Over millions of years, this organic material was affected by high temperatures and pressures, causing the oils and fats to be converted into hydrocarbons. This mixture of hydrocarbons is crude oil.

Crude oil has a low density, which means it could migrate upwards through the crust, where it often became trapped beneath impervious (unable to be passed through) rock (Figure 8.1.1). Accumulation of oil and gas under the rock creates an oil field.

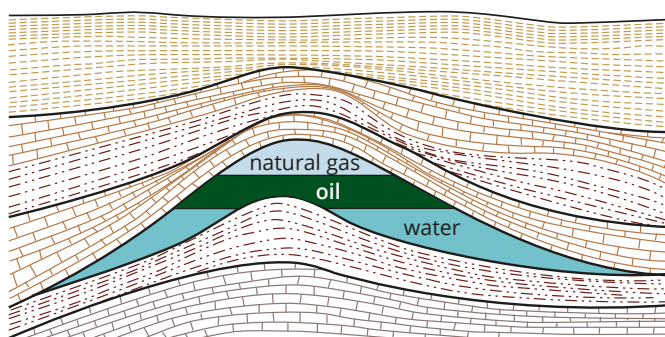


FIGURE 8.1.1 Typical structure of the impervious rock that traps oil and natural gas underground and creates an oil field

Crude oil is not used in its raw state. It is transported from oil fields to oil refineries where it undergoes **fractional distillation** in a fractionating tower as shown in Figure 8.1.2. In this process, the crude oil is separated into its various components, or fractions. Each fraction is made up of a range of hydrocarbons with similar boiling points and hence molecular masses. The components of crude oil that are obtained by fractional distillation are used for a wide range of purposes, but presently over 90% of them are used for fuels.

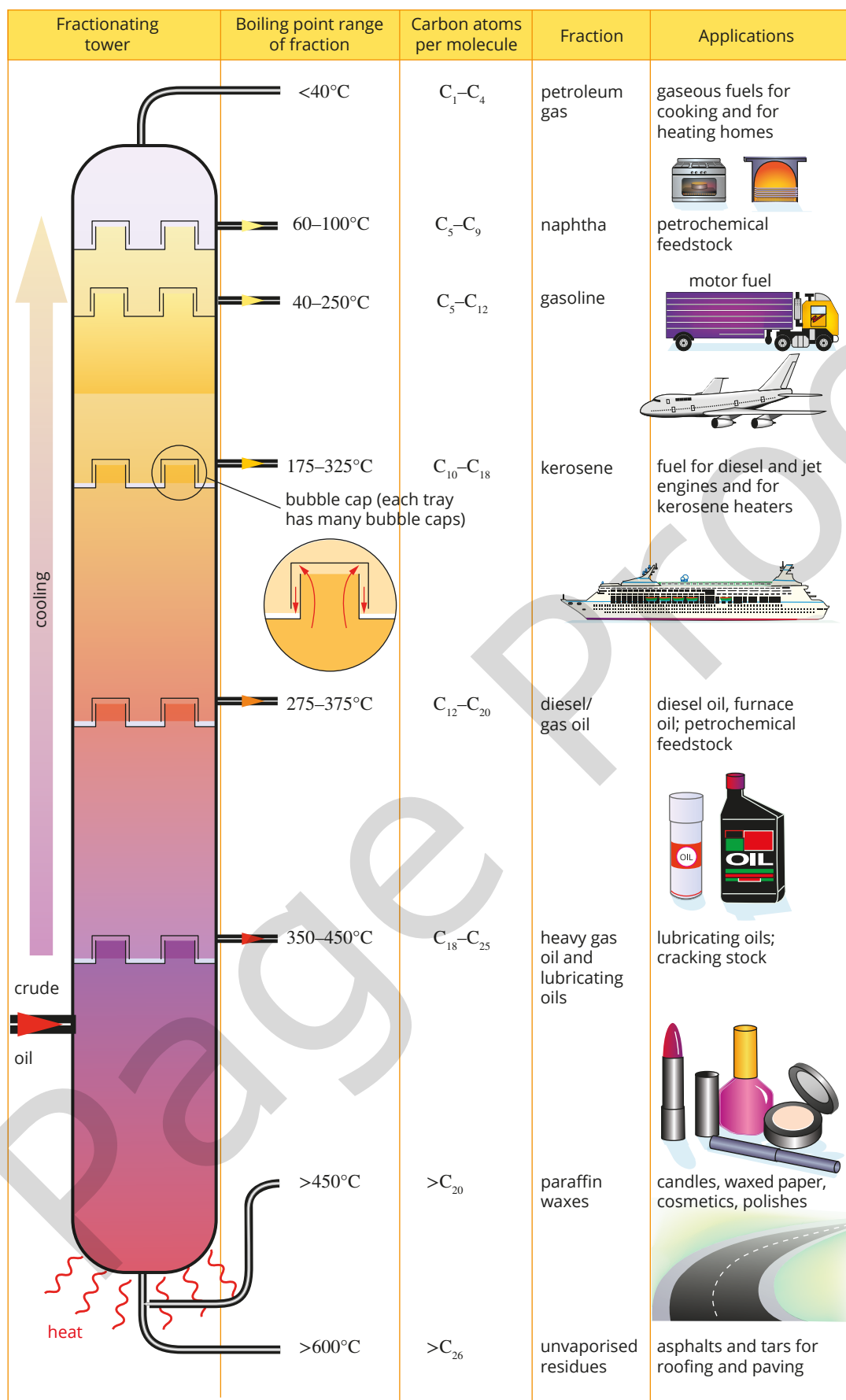


FIGURE 8.1.2 Fractions found in crude oil

Further processing of the crude oil fractions is needed to make these components even more useful.

Heavier fractions undergo a process called **cracking**, which is shown in Figure 8.1.3. This process breaks the larger hydrocarbon molecules into smaller molecules using heat and a catalyst. The smaller molecules and especially those with carbon–carbon double bonds (called alkenes) are needed by the petrochemical industry.

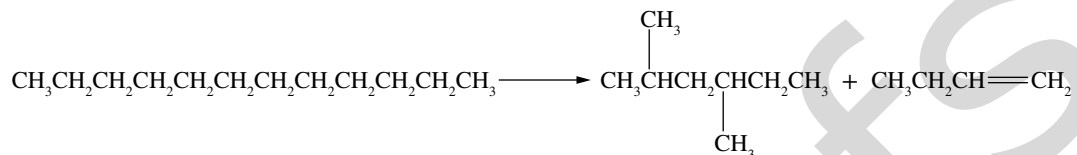


FIGURE 8.1.3 Cracking of a larger hydrocarbon molecule results in two smaller molecules, one of which has a carbon–carbon double bond.

ALKANES

Hydrocarbons are composed only of carbon and hydrogen. The simplest hydrocarbon is methane. In methane, one carbon atom is covalently bonded with four hydrogen atoms to form a molecule with the formula CH_4 .

Methane is a colourless, odourless gas that is very flammable. It is the main component of natural gas, which is used in many Australian homes for heating and cooking.

Hydrocarbons can be classified into several groups or series. Methane is the first of a series of compounds known as the **alkanes**. Alkanes are hydrocarbons that contain only single bonds. All the carbon–carbon bonds in alkanes are single covalent bonds. Molecules such as this are said to be **saturated**.

Each member of the alkane series differs from the previous member by a $-\text{CH}_2-$ unit. A series of molecules in which each member differs by $-\text{CH}_2-$ from the previous member is known as a **homologous series**.

Compounds that are members of the same homologous series have:

- a similar structure
- a pattern to their physical properties
- similar chemical properties
- the same general formula.

Alkanes have only weak forces between the molecules because they are ‘non-polar’. You will learn more about polar and non-polar molecules in Chapter 12. As the number of carbons and the size of the molecules increase within this homologous series, the strength of these forces increases, so the melting and boiling points of the alkanes increase.

Table 8.1.1 shows the first three members of the alkane series. Note that the bonds around each carbon atom adopt a tetrahedral shape and the boiling points increase as the molecules become larger.

The naming of the members of each homologous series follows set conventions. The **stem name**, or parent name, in propane (prop-) indicates there are three carbon atoms in the molecule. This same stem name (prop-) is used in other series for the member that contains three carbon atoms. The stem names used for molecules with between one and ten carbon atoms are listed in Table 8.1.2.

Alkanes are named by adding -ane after the stem name. For example, an alkane that contains eight carbon atoms is called octane. The molecular formula of octane is C_8H_{18} . The alkanes have the general formula $\text{C}_n\text{H}_{2n+2}$, where n stands for the number of carbon atoms. If an alkane molecule has 12 carbon atoms, the number of hydrogen atoms is $2n + 2 = 2 \times 12 + 2 = 26$. The molecular formula is therefore $\text{C}_{12}\text{H}_{26}$.

TABLE 8.1.1 Structure, properties and some uses of the first three alkanes

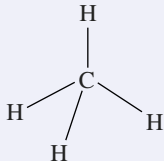
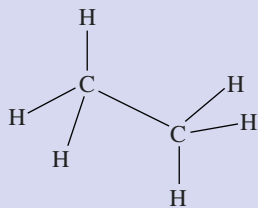
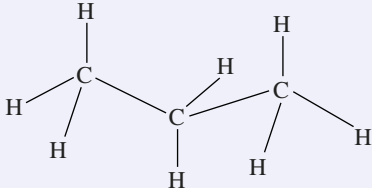
Name and molecular formula	Structural formula	Properties	Uses
methane, CH ₄		non-polar, gas, boiling point (BP) -164°C	cooking, Bunsen burners, gas heating
ethane, C ₂ H ₆		non-polar, gas, BP -87°C	conversion to ethene
propane, C ₃ H ₈		non-polar, gas, BP -42°C	liquid petroleum gas (LPG)

TABLE 8.1.2 Stem names used for molecules with between one and ten carbon atoms

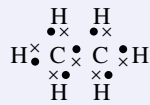
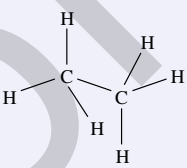
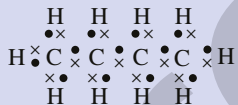
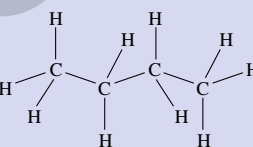
Stem (parent) name	Number of carbon atoms
meth-	1
eth-	2
prop-	3
but-	4
pent-	5
hex-	6
hept-	7
oct-	8
non-	9
dec-	10

i The name of an alkane ends in -ane and alkanes have the general formula C_nH_{2n+2} .

Writing formulae of alkanes

Sometimes you want to show more information about a molecule than just its overall composition. You can use a variety of ways to write the formulae of carbon compounds. Table 8.1.3 shows different ways of representing ethane and butane.

TABLE 8.1.3 Different ways of representing alkanes

Alkane	Molecular formula	Lewis structure diagram (Electron diagram)	Condensed structural formula (semistructural formula)	Structural formula
ethane	C_2H_6		CH_3CH_3	
butane	C_4H_{10}		$CH_3CH_2CH_2CH_3$ or $CH_3(CH_2)_2CH_3$	

Structural formulae show all the bonds in a molecule, but lone pairs can be omitted. **Condensed structural formulae** are also known as semistructural formulae and show the atoms that are connected to each carbon atom, but do not show the bonds.

Representing three-dimensional molecules on a two-dimensional computer screen or page is a challenge for chemists. For example, each carbon in an alkane has four bonds in a tetrahedral arrangement. When the shape of the molecule is not important, it is often drawn in a simplified two-dimensional format as shown in Figure 8.1.4.

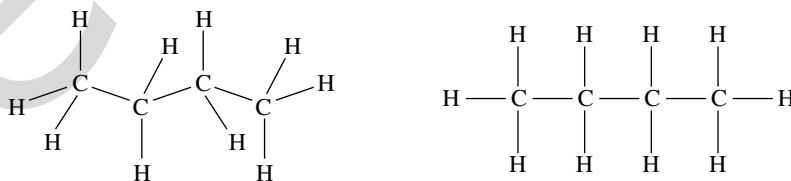


FIGURE 8.1.4 Molecules such as butane are often drawn in two dimensions for simplicity.

Structural isomers of alkanes

There is only one molecule that can be formed with the molecular formula of methane (CH_4). This is also the case for ethane (C_2H_6) and propane (C_3H_8). However, alkanes that have four or more carbon atoms have more than one possible structure.

Figure 8.1.7 shows two different molecules that have the molecular formula C_4H_{10} . Molecules that have the same molecular formula but have different arrangements of atoms are said to be **structural isomers** of each other. The more atoms in the molecule, the more possible isomers there are.

CHEMFILE

Have you got the patience for slow science?

The longest running laboratory experiment in the world was set up in 1927 at the University of Queensland in Brisbane. The long-running 'pitch drop' experiment is designed to show that pitch, the name given to a mixture of highly viscous hydrocarbons, is actually a liquid. Pitch, otherwise known as bitumen (Figure 8.1.5), appears solid on first inspection and shatters if you strike it with a hammer.

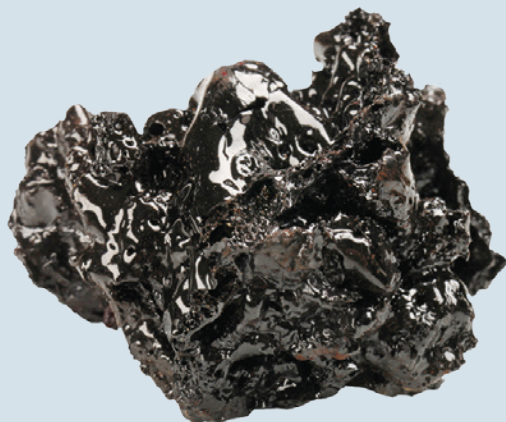


FIGURE 8.1.5 A sample of pitch at room temperature appears solid and shatters when hit with a hammer.

Professor Thomas Purnell set up the pitch experiment as shown in Figure 8.1.6 by placing a sample of pitch into a funnel encased in a sealed container. He wanted to demonstrate to students that pitch was indeed a liquid—just a very, very viscous one.

Over time, the pitch has settled and slowly flowed through the funnel producing a drop on average once every decade. The ninth drop was recorded in April of 2014.

Today bitumen is used extensively, forming the main component of road surfaces. Under normal conditions, the road surface appears solid. However, heating reduces the viscosity and road workers are able to work the pliable bitumen.



FIGURE 8.1.6 Professor Thomas Purnell set up the famous University of Queensland pitch drop experiment in 1927. Professor John Mainstone, shown here, oversaw the experiment for many years.

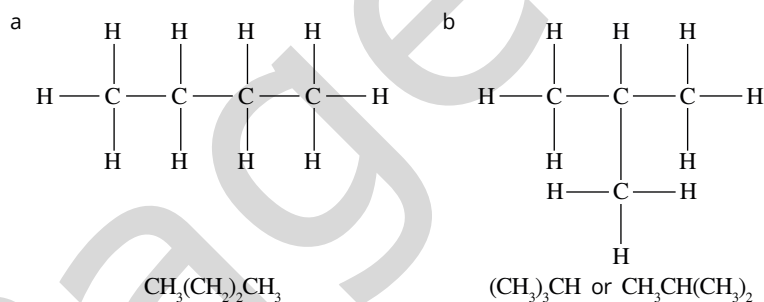


FIGURE 8.1.7 There are two structural isomers that have the molecular formula C_4H_{10} .

Figure 8.1.7a shows the straight-chain isomer of butane, so named because the four carbon atoms are bonded in a continuous chain. Figure 8.1.7b is the branched isomer because you could consider it as a straight-chain alkane, with three carbon atoms in the chain, with a $-\text{CH}_3$ **side group** attached. A $-\text{CH}_3$ side group is called a methyl group, as its structure is similar to that of methane with one less hydrogen. A $-\text{CH}_2\text{CH}_3$ side group is called an ethyl group.

Therefore, Figure 8.1.7b is named methylpropane, as it can be regarded as a propane molecule with a methyl group attached. Methyl and ethyl groups are examples of **alkyl groups**. Alkyl groups have one less hydrogen atom than the corresponding alkane of the same name, so the general formula of an alkyl group is $-\text{C}_n\text{H}_{2n+1}$.

Table 8.1.4 lists the names of the alkyl groups with 1–5 carbon atoms.

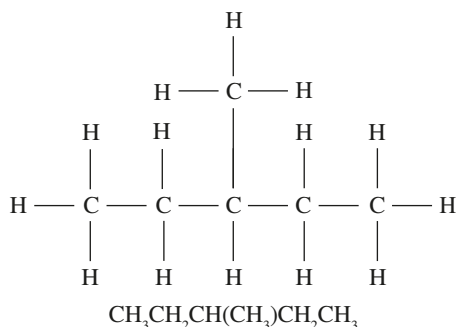


FIGURE 8.1.8 The structural and condensed structural formula of a branched alkane

TABLE 8.1.4 Names of the alkyl groups with 1–5 carbons

Alkyl group	Corresponding alkane	Name of alkyl group
$-\text{CH}_3$	methane	methyl
$-\text{CH}_2\text{CH}_3$	ethane	ethyl
$-\text{CH}_2\text{CH}_2\text{CH}_3$	propane	propyl
$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	butane	butyl
$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	pentane	pentyl

When writing the condensed structural formula of a branched alkane, alkyl groups are written in brackets (Figure 8.1.8).

Structural isomers are different compounds with different physical and chemical properties. Table 8.1.5 shows the isomers of hexane and their respective melting points and boiling points. The boiling and melting points are different because the forces of attractions between the different versions of hexane molecules are affected by the structure of the different molecules (isomers).

TABLE 8.1.5 Melting and boiling points of the isomers of hexane

Isomer	Melting point (°C)	Boiling point (°C)
$\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$ hexane	−95.3	68.7
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{CH}_2 - \text{CH} - \text{CH}_2 - \text{CH}_3 \end{array}$ 3-methylpentane	−118.0	63.3
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH} - \text{CH}_3 \end{array}$ 2-methylpentane	−153.7	60.3
$\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{CH}_3 - \text{CH} - \text{CH} - \text{CH}_3 \end{array}$ 2,3-dimethylbutane	−128.6	58
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{CH}_2 - \text{C} - \text{CH}_3 \\ \\ \text{CH}_3 \end{array}$ 2,2-dimethylbutane	−99.8	49.7

CHEMFILE

Burning ice

Large amounts of methane gas are stored around the world trapped as solid methane hydrate. These methane hydrates look very similar to ice with one main difference—they burn.

The burning ice is a result of the methane gas being released from the crystal structure of the ice as it melts (Figure 8.1.9).

Large deposits of methane hydrate are trapped in the Arctic regions of the Earth and deep in the oceans where low temperatures and high pressures trap methane. This source of methane is produced by the anaerobic decomposition of organic material by bacteria. The lattice structure of methane hydrate is shown in Figure 8.1.10.



FIGURE 8.1.9 The combustion of blocks of methane hydrate gives the appearance of burning ice. As the ice melts, methane gas is released from the lattice structure, providing the fuel for the combustion reaction.

Oil companies have known of the existence of methane hydrate as far back as the 1930s. However it is only now, with growing concerns over dwindling fossil fuel supplies and global warming, that scientists are endeavouring to find a way to safely use these large fuel sources.

One fear is that with rising ocean temperatures the deposits of methane hydrate throughout the oceans will melt, releasing large amounts of methane into the atmosphere. Methane is a major contributor to the enhanced greenhouse effect, so a release of this size is likely to accelerate the effects of global climate change.

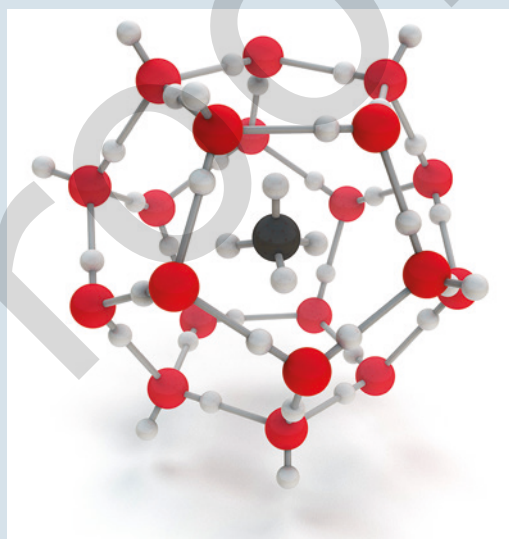


FIGURE 8.1.10 Structure of methane hydrate. Methane molecules are trapped inside the hexagonal structure of ice.

NAMING ALKANES

The systematic naming of carbon compounds is controlled by the International Union of Pure and Applied Chemists (IUPAC). Under the IUPAC system, the name of the compound provides details of its structure. The following rules apply when naming alkanes.

- 1 Identify the longest unbranched carbon chain and count the number of carbon atoms in that chain. This gives the molecule its stem name.
- 2 If there are branches to be added to the name, number the carbon atoms in the chain from the end of the chain that will give the smallest numbers to branching groups.
- 3 Identify the type and the number of the alkyl groups attached to the chain.
- 4 Place the number and position of each of the alkyl groups at the beginning of the compound's name.
- 5 If two identical side chains are present, use 'di-' as a prefix; for three use 'tri-'.
- 6 If there are different alkyl side chains, list them in alphabetical order at the start of the name, with their numbers to indicate their respective positions.

i Carefully check that you have identified the longest unbranched carbon chain. Sometimes the longest carbon chain is not drawn in a straight line.

The following steps show the process of naming the isomer of hexane shown in Figure 8.1.11.

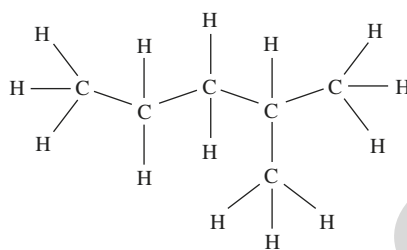
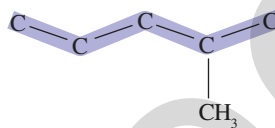
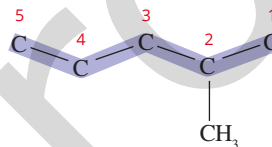


FIGURE 8.1.11 The IUPAC system will be used to name this isomer of hexane.

- 1 Identify the longest unbranched carbon chain.

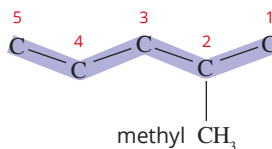


- 2 Number the carbons, starting from the end closest to the branch.



- 3 Name the side branches and main chain.

5 carbon atoms = pentane



- 4 Combine all components to write the full name: 2-methylpentane.

Figure 8.1.12 shows examples of applications of these rules. Note the use of the prefix 'di-' to indicate the presence of two methyl side branches and the numbering to indicate their position along the longest continuous carbon chain (for example, 3,3- indicates that both side branches come off the third carbon atom).

Note that in the names, numbers are separated by commas, numbers are separated from letters with a hyphen. You also only need to use numbers when there is more than one possible position on the chain for the groups. For example, you need to include the '2' in 2-methylpentane because the methyl group could be on the second or the third carbon atom. But for methylpropane, no number is required as the only possible place for the methyl group to be attached to the chain is the second carbon atom.

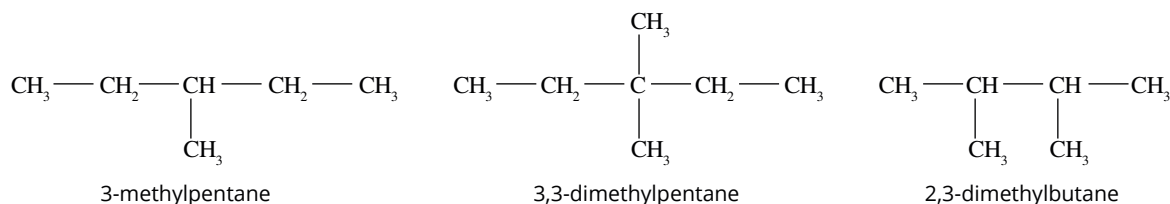
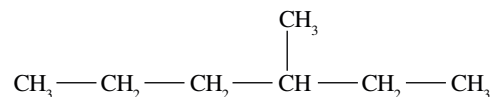


FIGURE 8.1.12 IUPAC systematic names for three alkanes. Note that there are no spaces in the names of these compounds.

Worked example 8.1.1

IUPAC NAMING SYSTEM FOR ALKANES

Write the systematic name for the following molecule.

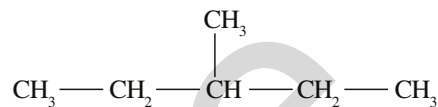


Thinking	Working
Identify the longest carbon chain in the molecule. The stem name of the molecule is based on this longest chain.	There are six carbons in the longest chain. The stem name is based on hexane.
Number the carbon atoms starting from the end closest to the branch.	$\begin{array}{ccccccc} & & & \text{CH}_3 & & & \\ & & & & & & \\ \text{CH}_3 & - & \text{CH}_2 & - & \text{CH}_2 & - & \text{CH} & - & \text{CH}_2 & - & \text{CH}_3 \\ \leftarrow & 6 & 5 & 4 & 3 & 2 & 1 \end{array}$
Identify the branch.	The side chain is a methyl group.
Combine all components.	The name of the molecule is 3-methylhexane.

Worked example: Try yourself 8.1.1

IUPAC NAMING SYSTEM FOR ALKANES

Write the systematic name for the following molecule.

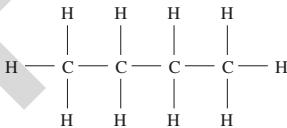
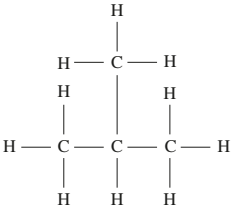
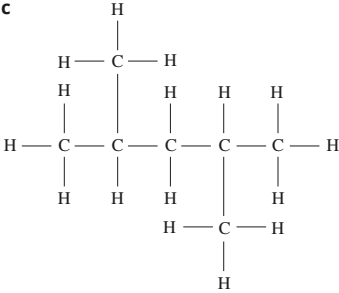
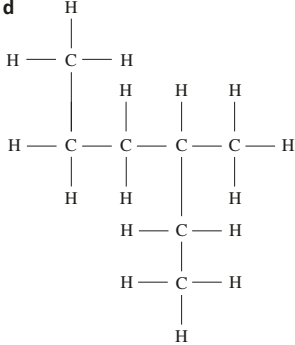


8.1 Review

SUMMARY

- Hydrocarbons are compounds containing carbon and hydrogen only.
- Crude oil is the source of many hydrocarbons. These hydrocarbons are separated from one another by fractional distillation.
- In a homologous series, each member has one more $\text{-CH}_2\text{-}$ unit than the previous member. Members of a homologous series have similar structures and chemical properties and the same general formula. Alkanes are an example of a homologous series.
- Alkanes are saturated hydrocarbon molecules that contain only single bonds and have a general formula $\text{C}_n\text{H}_{2n+2}$.
- Structural isomers are molecules with the same molecular formula but different arrangements of atoms.
- Hydrocarbon molecules can be drawn using structural formulae or condensed structural formulae.
- The IUPAC naming system is used to provide systematic names for hydrocarbon molecules. Names are based on the longest unbranched carbon chain.

KEY QUESTIONS

- From which one or more of the following sources does crude oil originate?
A plant material
B forest animals that have recently died
C dinosaurs only
D marine microorganisms
- Methane is the smallest hydrocarbon molecule.
 - What is the molecular formula of methane?
 - Why is methane a hydrocarbon?
 - Why does the carbon atom in a methane molecule bond to four, rather than two, three, five or any other number of hydrogen atoms?
 - Draw the structural formula of methane. Why does methane have this particular arrangement of hydrogen atoms around each carbon atom?
- Answer the following questions relating to the hydrocarbon with the molecular formula, C_3H_8 .
 - What is the name of this hydrocarbon?
 - Draw the structural formula of this hydrocarbon.
 - Write the condensed structural formula of this hydrocarbon.
- Write the systematic names of these alkanes based on their condensed structural formulae.
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$
 - $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$
 - $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}(\text{CH}_3)_2$
- Write the systematic names of the alkanes shown here based on their structural formulae.
 - 
 - 
 - 
 - 
- Draw the structural formulae of these alkanes based on their systematic names.
 - hexane
 - 3-methylhexane
 - 3,3-dimethylpentane
 - 3-ethyl-2-methylpentane

8.2 Alkenes

Carbon forms many compounds with hydrogen in which there are double or even triple bonds between the carbon atoms. These compounds are called **unsaturated** hydrocarbons. A double bond is formed when two pairs of electrons are shared and a triple bond is formed when three pairs of electrons are shared. Some of the most useful carbon compounds are unsaturated. This section focuses on the structures and naming of hydrocarbon compounds with carbon-carbon double bonds.

ALKENES

Alkenes are a homologous series of hydrocarbons with at least one carbon-carbon double bond. The general formula for the alkenes with one double bond is C_nH_{2n} . The presence of a carbon-carbon double bond means the alkenes are described as unsaturated molecules. Alkenes are more reactive than alkanes, which do not have any carbon-carbon double bonds.

i The name of an alkene ends in -ene and alkenes have the general formula C_nH_{2n} .

The simplest alkene is ethene (C_2H_4). The next member in the series is propene (C_3H_6), which has an additional $-CH_2-$ unit. The first three members of the alkene homologous series are shown in Table 8.2.1. Note that the angle of the bonds around a carbon atom with a double bond is 120° . All alkenes are non-polar and hence do not dissolve in water.

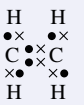
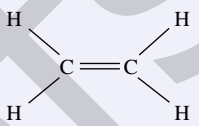
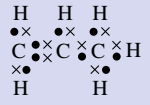
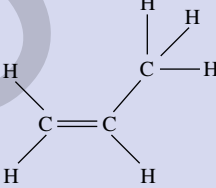
TABLE 8.2.1 Structure, properties and some uses of the first three alkenes

Name and molecular formula	Structural formula	Properties	Uses
ethene, C_2H_4		non-polar, gas, boiling point (BP) -78.4°C	in the manufacture of a wide range of chemicals
propene, C_3H_6		non-polar, gas, BP -47.7°C	in the manufacture of propene oxide and polymers
butene, C_4H_8		non-polar, gas, BP -6.3°C	in the manufacture of butanol and polymers

Writing formulae of alkenes

Like alkanes, there are a variety of ways for writing the formulae of alkenes. Table 8.2.2 shows different ways of representing ethene and propene.

TABLE 8.2.2 Different ways of representing alkenes

Alkene	Molecular formula	Electron dot diagram (Lewis structure)	Condensed structural formula (semistructural formula)	Structural formula
ethene	C_2H_4		CH_2CH_2	
propene	C_3H_6		CH_2CHCH_3	

Structural isomers of alkenes

For alkene molecules that contain more than three carbon atoms, structural isomers will exist. As you can see in Figure 8.2.1, isomers may result from branches in the carbon chain, or if the carbon–carbon double bond is in different positions.

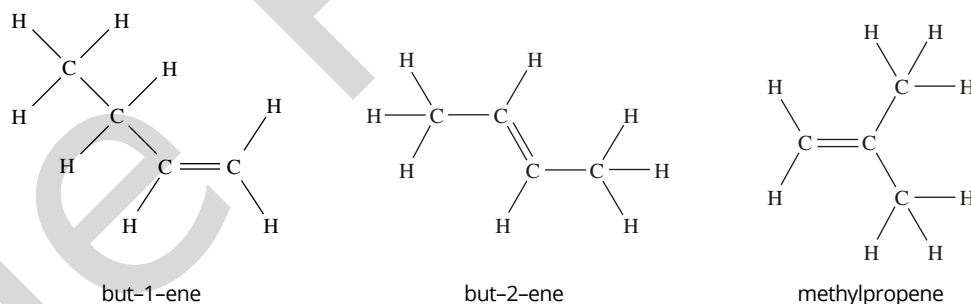


FIGURE 8.2.1 Isomers of butene. Isomers of alkenes can have branches in the carbon chain or the double bond at different points in the carbon chain.

When naming alkene isomers, the carbons are numbered from the end of the carbon chain that gives the lowest number to the first carbon in the double bond. The location of the double bond in the molecule is indicated by this number. In Figure 8.2.1, you can see that the double bond in but-1-ene is between carbons 1 and 2. The double bond in but-2-ene is between carbons 2 and 3.

Geometric (*cis-trans*) isomers of alkenes

In **geometric isomers** of alkenes, the atoms in the isomers are joined in the same order but have a different arrangement in space. This occurs because the atoms associated with a carbon–carbon double bond are fixed in position.

Two atoms joined by a single bond can rotate freely around the single bond. At first glance, it may look as if the atoms in the two molecules shown below have a different spatial arrangement. However, they are the same molecule because free rotation occurs around the single bond (Figure 8.2.2).

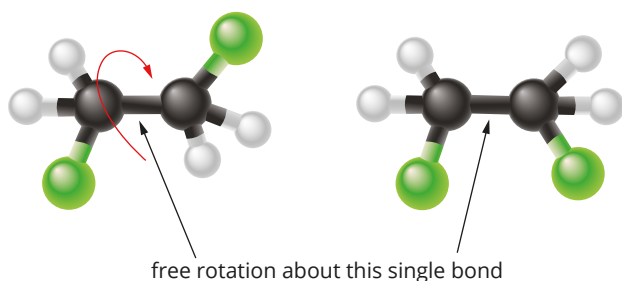


FIGURE 8.2.2 Atoms can freely rotate around a carbon–carbon single bond. These two models are representations of the same molecule.

On the other hand, two atoms that are joined by a double bond do not rotate freely because the double bond locks the atoms firmly in position. You can see that the models below represent two molecules that cannot be rotated to form the same structure (Figure 8.2.3).

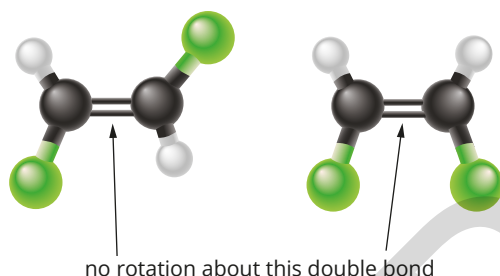


FIGURE 8.2.3 Atoms cannot freely rotate around a carbon–carbon double bond. These two models do not represent the same molecule.

Geometric, or *cis-trans* isomerism can occur when there are two different groups attached to each carbon atom involved in the double bond (Figure 8.2.4).

- In *cis* isomers the same group on each carbon is located on the same side of the double bond.
- In *trans* isomers the same group on each carbon is located on opposite sides of the double bond.

NAMING ALKENES

The following rules apply when naming alkenes.

- 1 Identify the longest unbranched carbon chain. For alkenes, the chain must include the double bond.
- 2 Number the carbon atoms in the chain from the end of the chain that will give the smallest numbers to the double-bonded carbon atoms.
- 3 Name the alkyl groups after the alkane from which they are derived.
- 4 Identify the position of the double bond by the number of the first carbon atom involved in the bond. Use the suffix ‘-ene’ to indicate the presence of a double bond, e.g. hex-2-ene.
- 5 List the number and position of each of the alkyl groups at the beginning of the compound’s name in alphabetical order.
- 6 If two identical side chains are present, use ‘di-’ as a prefix; for three use ‘tri-’.
- 7 Identify whether the structure exhibits geometric isomerism and if so, whether it is a *cis* or *trans* isomer.

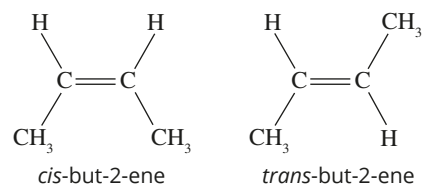


FIGURE 8.2.4 *Cis* and *trans* isomers of but-2-ene. The two molecules have the same condensed structural formula, but they have different structures due to a different spatial arrangement of atoms.

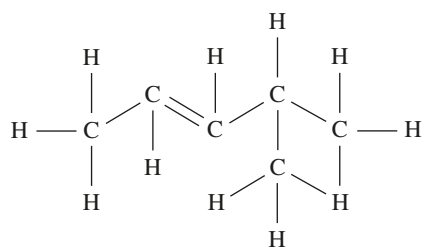
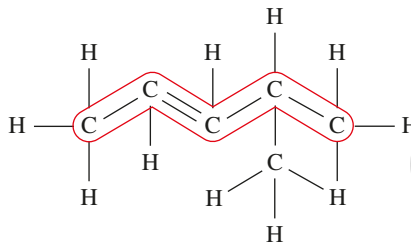


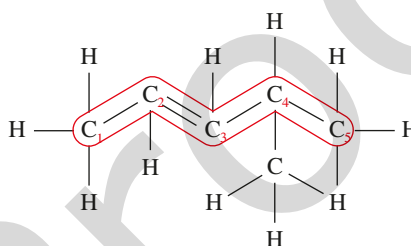
FIGURE 8.2.5 The IUPAC system will be used to name this isomer of hexene.

The following steps show the process of naming the isomer of hexene shown in Figure 8.2.5.

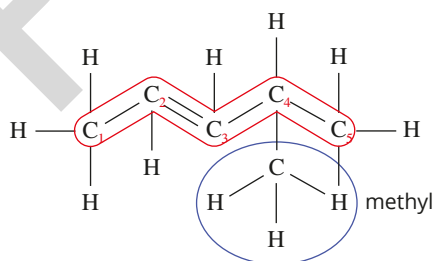
- 1 Identify the longest carbon chain that contains the double bond. The name of the molecule is based on this chain.



- The longest chain contains five carbons—pentene
- 2 Number the carbons, starting from the end closest to the double bond. Note the position of the double bond.



- It is a 5-carbon chain with the double bond starting at C number 2.
- 3 Name each side branch and find the number of the carbon that it is on.



- There is a methyl group on C number 4.
- 4 Combine all components to write the full name.
4-methylpent-2-ene
 - 5 Identify whether the structure exhibits geometric isomerism and, if so, whether it is a *cis* or *trans* isomer.
 - It is a *trans* isomer
 - *trans*-4-methylpent-2-ene

CHEMFILE

How do we see? *Cis-trans* isomerism of retinal

In 1967, Georges Wald was awarded a share of the Nobel Prize for Physiology and Medicine for his work with pigments in the retina. He discovered that vitamin A (retinal) was critical to our vision and investigated the way in which it absorbs light.

Our vision begins when molecules of retinal absorb a small packet of light called a photon. This absorption causes the *cis*-form of retinal to be converted to the *trans*-form (Figure 8.2.6), changing the shape of the retinal molecule.

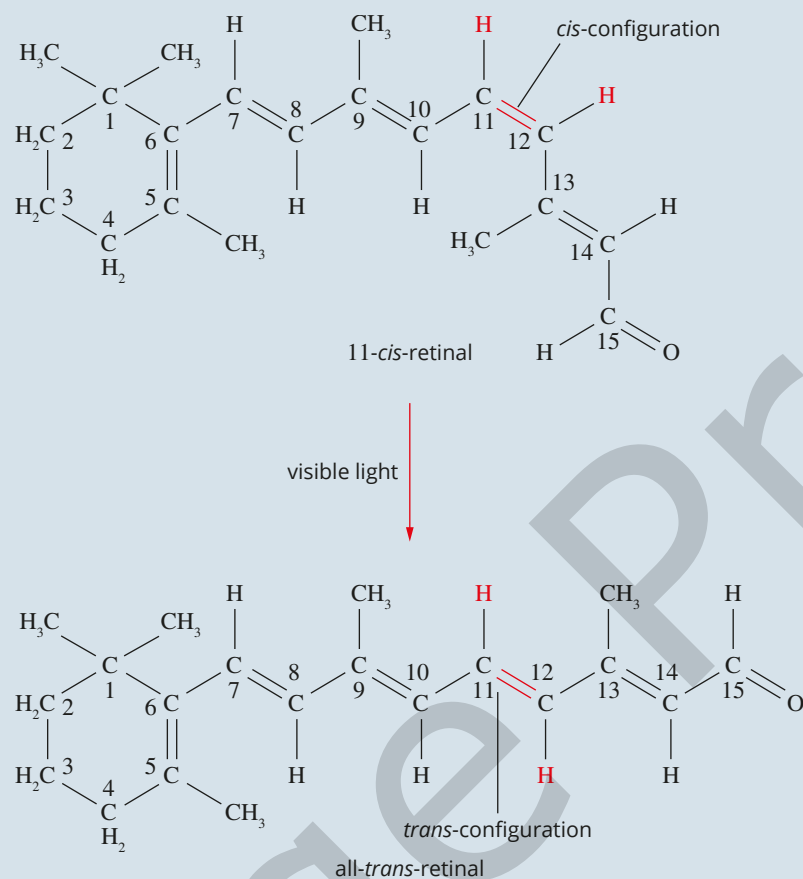


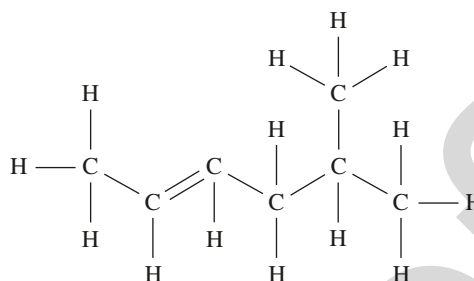
FIGURE 8.2.6 When *cis*-retinal absorbs a photon of light, it is converted to the *trans* isomer.

The *cis* form of retinal is able to bond tightly to the protein opsin in our eyes. After it has reacted with light and been converted to the *trans* form, it no longer fits closely within the protein, as its shape has changed. This causes the opsin protein to change its shape, and begins a signalling pathway that sends an impulse along the optic nerve to our brains. This is how the light that enters our eyes is converted to vision in our brains.

Worked example 8.2.1

IUPAC NAMING SYSTEM FOR ALKENES

Write the systematic name for the following molecule.



Thinking

Identify the longest carbon chain in the molecule that contains the double bond.
The name of the molecule is based on this longest chain.

Identify whether the compound exhibits geometric isomerism.

Number the carbon atoms starting from the end closest to the double bond.
Note the position of any double bond.

Identify each branch and the number carbon that it is on.

Combine all components.

Working

There are six carbons in the longest chain.

The name is based on hexene.

It is a *cis* isomer.

There is a double bond on carbon number 2, so the longest chain is hex-2-ene.

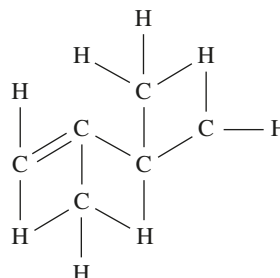
The side chain is a methyl group and it is on carbon number 5.

The name of the molecule is *cis*-5-methylhex-2-ene.

Worked example: Try yourself 8.2.1

IUPAC NAMING SYSTEM FOR ALKENES

Write the systematic name for the following molecule.



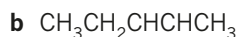
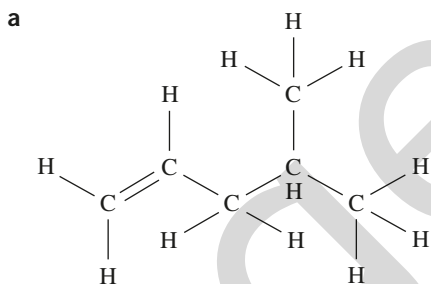
8.2 Review

SUMMARY

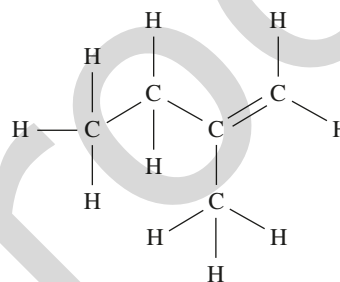
- Unsaturated hydrocarbons are compounds containing carbon and hydrogen only, with a carbon–carbon double bond.
- Alkenes have the general formula C_nH_{2n} .
- Names are based on the longest unbranched carbon chain. The position of the double bond determines which end to start numbering from.
- Alkenes can demonstrate structural isomerism and geometric (*cis-trans*) isomerism

KEY QUESTIONS

- List the following alkenes in order from lowest to highest number of atoms: pentene, ethene, methylpropene, octene, propene.
- How many carbon atoms are in methylpropene?
 - How many hydrogen atoms are in methylpropene?
 - Of which straight chain alkene is methylpropene an isomer?
- Which of the following is the correct condensed structural formula of pent-1-ene?
A C_5H_{12}
B $CH_3CH_2CH_2CHCH_2$
C $CH_3CH_2CH_2CCH$
- Write the systematic names of the unsaturated hydrocarbons shown below, based on their formulae.



c



- Draw the structural formulae of these unsaturated hydrocarbons based on their systematic names.
 - cis*-but-2-ene
 - hex-1-ene
 - 3-methylpent-1-ene
 - 4,4-dimethylpent-1-ene

8.3 Benzene

In 1825, Michael Faraday isolated a new hydrocarbon, named benzene (C_6H_6). The structure and properties of benzene puzzled scientists for 40 years. Although its molecular formula was found to be C_6H_6 , scientists were unable to draw a structure that matched the observed properties of benzene (Figure 8.3.1).



FIGURE 8.3.1 These are two molecular structures with the formula C_6H_6 . Neither of these structures accounts for the chemical properties of benzene.

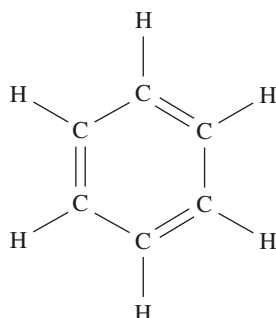


FIGURE 8.3.2 The Kekulé structure for benzene is a ring of six carbon atoms, joined by alternating single and double bonds. This structure was later found to be inconsistent with its chemical reactivity.

Benzene was found to be unexpectedly stable. August Kekulé made a major contribution to the puzzle in 1865. He proposed a ring structure in which alternate carbon atoms were joined by double bonds (Figure 8.3.2).

In his own words, Kekulé described a dream that led him to the discovery of benzene's structure:

'Again the atoms were gambolling before my eyes. This time the smaller groups kept modestly in the background. My mental eye, rendered more acute by repeated visions of this kind, could now distinguish larger structures of manifold conformation, long rows, sometimes more closely fitted together, all twining and twisting in snakelike motion. But look! What was that? One of the snakes had seized hold of its own tail and the form whirled mockingly before my eyes. As if by a flash of lighting I awoke.'

It is now known that benzene has a ring structure, but it does not have alternating double and single bonds. The electrons that make up the double bonds are not in fixed positions: they are delocalised and can move through the ring. In fact, the carbon-to-carbon bonds are all the same length and the molecule is more stable than expected of an unsaturated molecule (Figure 8.3.3).

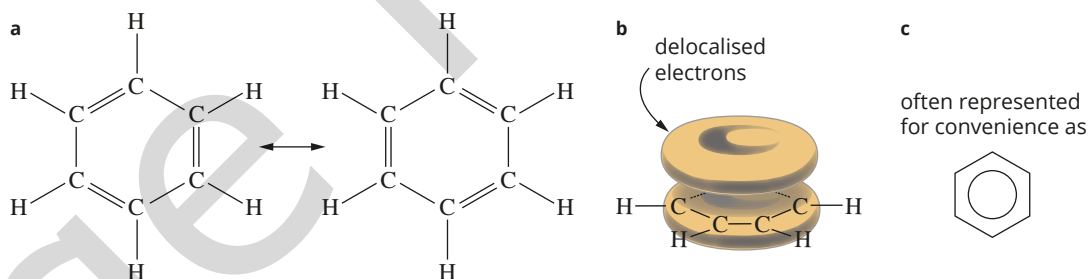


FIGURE 8.3.3 Three representations of a benzene molecule; (a) as an alternating combination (resonance) of two 'Kekulé structures', (b) visualising the delocalised electrons and (c) the convenient way the structural formula can be written.

Benzene is one of the components of petrol. It was once widely used as a solvent but, because it is carcinogenic (increases the risk of cancer), it has been phased out of this role. Polystyrene, phenol and benzoic acid are examples of the many commercial products that are manufactured from reactions involving benzene.

Molecules such as benzene, which can be represented as containing a ring of carbon atoms with alternating single and double bonds, are referred to as aromatic hydrocarbons, a name that comes from the fact that many of them have sweet or pleasant odours. The term 'arenes' is often used for aromatic hydrocarbons.

CHEMISTRY IN ACTION

Mothballs in space

Mothballs traditionally contained the active ingredient naphthalene. Naphthalene consists of a fused pair of benzene rings and has the formula $C_{10}H_8$ (Figure 8.3.4). It was first discovered in the early 1820s, when a white solid with a pungent odour was distilled from coal tar. It has been used as a household fumigant, as it is toxic to the moths that attack fabric. Indeed, although mothballs now contain the safer compound 1,4-dichlorobenzene, naphthalene was used for many years for this purpose. Its effectiveness as a repellent is such that it is produced in trace amounts by termites, which use it to repel ants and poisonous fungi. Unfortunately, naphthalene is now known to carry health risks. It is carcinogenic to humans and animals and exposure to large amounts may damage or destroy red blood cells.

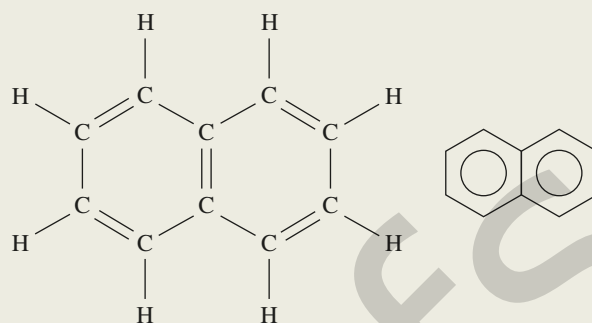


FIGURE 8.3.4 Two different representations of the compound naphthalene. Naphthalene consists of two fused benzene rings.

Interestingly, naphthalene has also been detected in space, having been observed by astronomers as part of the spectrum of Unidentified Infrared Emissions. In space, it is protonated naphthalene (containing an extra proton), that is responsible for a large part of the infrared emissions. It is estimated that more than 20% of all the carbon in the universe might be associated with hydrocarbons such as naphthalene that contain multiple benzene rings.

8.3 Review

SUMMARY

- Benzene has the formula C_6H_6 .
- Benzene consists of a 6-sided carbon ring, with 6 delocalised electrons contained within the ring. All C–C bonds are of the same length, intermediate between a single and double C–C bond.

KEY QUESTIONS

- Which one of the following best describes the bonds between carbon atoms in benzene?
A All are single bonds.
B All are double bonds.
C They are alternating single and double bonds.
D They are intermediate between single and double bonds.
- In 1865, August Kekulé proposed a planar structure for benzene in which the carbon atoms were bonded in a hexagonal ring, with alternating double and single bonds. Which of the following best explains why his description of the structure is incorrect?
A The carbon atoms adopt a circular arrangement, rather than a hexagon.
B A benzene molecule contains only double bonds between the carbon atoms.
C One electron from each carbon is free to move around the ring.
D There are three adjacent double bonds in a benzene molecule.

8.4 Reactions of hydrocarbons

Hydrocarbons take part in a number of different types of reactions. All hydrocarbons undergo combustion, reacting with oxygen gas in exothermic reactions. For this reason, they are very useful as fuels. Alkanes and benzene also undergo substitution reactions and unsaturated alkenes, being more reactive than saturated hydrocarbons, undergo addition reactions.

Substitution reactions

Alkanes can react with halogens, such as chlorine and bromine, in the presence of ultraviolet (UV) light. In these reactions, a carbon–hydrogen bond is broken and replaced by a carbon–halogen bond. This type of reaction is called a substitution reaction. An example of a substitution reaction is shown below in Figure 8.4.1.

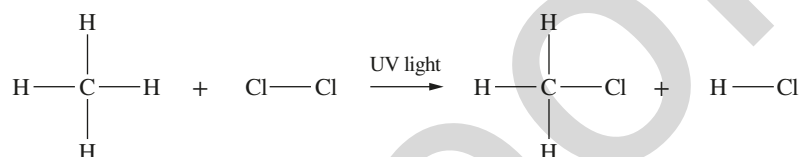


FIGURE 8.4.1 Methane reacts with chlorine gas. A chlorine atom is substituted for a hydrogen atom on the methane molecule.

If an excess of the halogen is present (i.e. **excess reactant**) and the reaction is allowed to proceed to completion, further substitution reactions can take place. For example, when there is a large excess of chlorine present relative to methane, the favoured product is tetrachloromethane, with each of the four hydrogen atoms having been substituted for chlorine atoms (Figure 8.4.2).

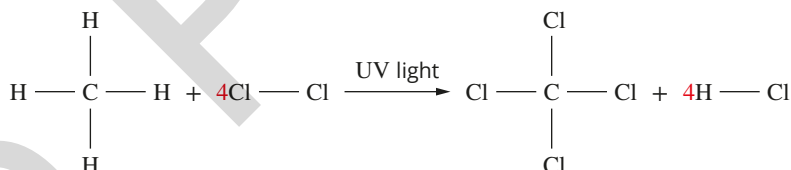


FIGURE 8.4.2 When methane reacts with an excess of chlorine gas, four chlorine atoms are substituted for hydrogen atoms on the methane molecule.

Benzene also undergoes substitution reactions in which one of the hydrogen atoms is replaced by another atom or a group of atoms. Alkane substitution reactions require UV light, however, substitution reactions of benzene also use catalysts, such as aluminium chloride or aluminium bromide, as indicated in Figure 8.4.3 below.

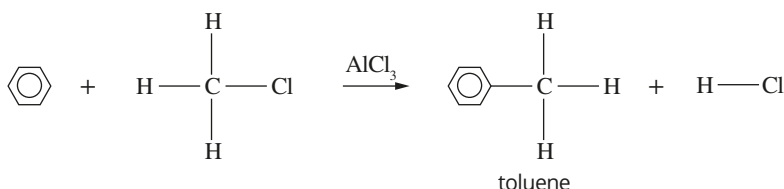
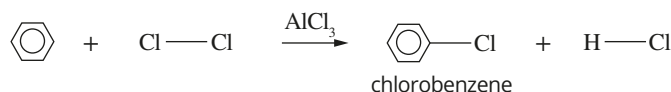
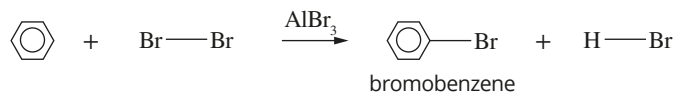


FIGURE 8.4.3 Benzene undergoes catalysed substitution reactions which involve hydrogen atoms being substituted for Br, Cl atoms or a methyl (–CH₃) group.

i Alkanes and benzene undergo substitution reactions, normally in the presence of UV light. In these reactions, a hydrogen atom is substituted for another atom or group of atoms.

Addition reactions

The double carbon–carbon covalent bond in alkenes has a significant effect on the chemical properties of this homologous series. Alkenes are generally more reactive compounds, reacting more readily and with more chemicals than alkanes, which contain only single bonds. The reactions of alkenes usually involve addition of a small molecule to the double bond of the alkene. These reactions are called **addition reactions**.

During addition reactions:

- two reactant molecules combine to form one product molecule
- the carbon–carbon double bond ‘opens up’ to become a single bond
- an unsaturated compound becomes saturated
- the atoms of the small molecule adding to the alkene are ‘added across the double bond’, so that one atom or group from the molecule forms a bond to the carbon atoms on each end of the double bond.

Unlike substitution reactions, there is no inorganic product formed. All the atoms in the reactants end up in the final product.

Four different types of addition reactions of alkenes are explained in detail below.

Reaction of alkenes with hydrogen

Alkenes react with hydrogen gas in the presence of a metal catalyst, such as nickel, to form alkanes. This reaction is known as a hydrogenation reaction and forms a saturated alkane. The reaction shown in Figure 8.4.4 is hydrogenation of ethene with hydrogen gas to produce ethane.

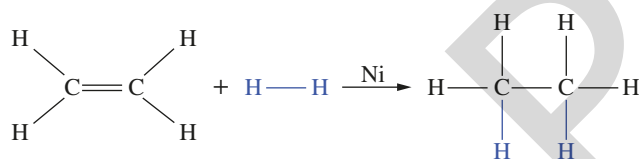


FIGURE 8.4.4 Addition reaction of ethene with hydrogen to form the corresponding saturated alkane, ethane

The reaction is too slow for the reaction to proceed at room temperature without a catalyst.

Reaction of alkenes with halogens

Figure 8.4.5 shows the reaction of ethene with bromine. The halogen adds across the double bond of the molecule, so in the product there is one bromine atom attached to each carbon atom.

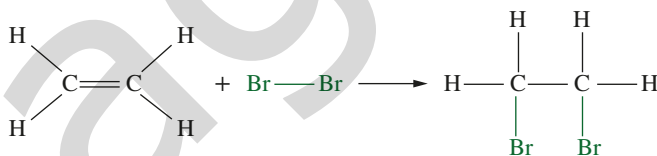


FIGURE 8.4.5 Addition reaction of ethene with bromine

This reaction proceeds at room temperature without a catalyst. Other halogens such as Cl_2 and I_2 also undergo addition reactions with alkenes.

i Alkenes undergo addition reactions. In these reactions, the double carbon–carbon bond is broken and a new atom or group of atoms is added to each carbon atom.

Reactions of alkenes with hydrogen halides

Alkenes also react with hydrogen halides (HCl, HBr, HF, HI) by addition reactions. Figure 8.4.6 shows the reaction of but-2-ene with hydrogen chloride to produce a single product. In this reaction, a hydrogen atom adds to one of the carbon atoms in the carbon-carbon double bond and the chlorine atom adds to the other carbon atom.

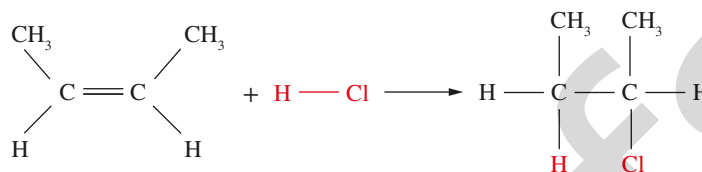


FIGURE 8.4.6 The addition reaction of but-2-ene with hydrogen chloride

However, as Figure 8.4.7 shows, when hydrogen chloride is reacted with but-1-ene, two products are possible.

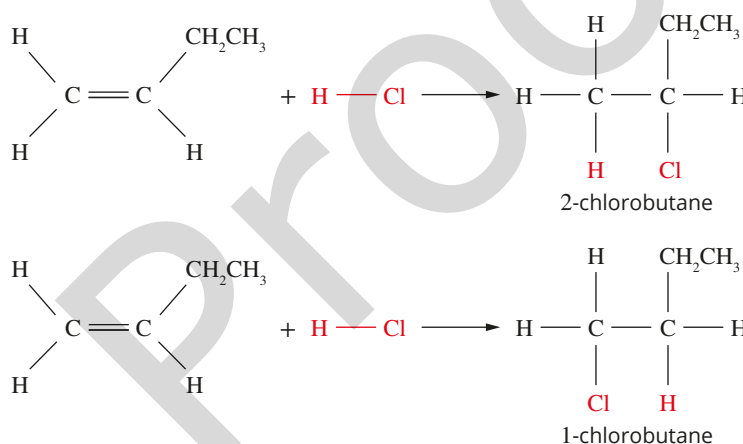


FIGURE 8.4.7 Addition reaction of but-1-ene with hydrogen chloride. Two isomers are possible as products.

The addition reaction can produce two isomers. In one product, the hydrogen atom from the hydrogen chloride molecule has been added to the carbon atom in the carbon-carbon double bond at the end of the but-1-ene molecule (C1). In the other product, the hydrogen atom has been added to a carbon atom at the other end of the carbon-carbon double bond (C2).

When you react an asymmetrical alkene such as but-1-ene with an asymmetrical reactant, isomers are produced. More of one isomer is usually produced than the other. The reasons for this are beyond the scope of this course (see the Extension below).

Reactions of alkenes with water

Alkenes react with water under specific conditions to form the corresponding alcohol. For example, ethanol can be produced by an addition reaction of ethene and water, using a catalyst to increase the rate of the reaction. Figure 8.4.8 shows the addition reaction of steam and ethene, using a phosphoric acid catalyst. This reaction is used extensively in industry for the production of ethanol.

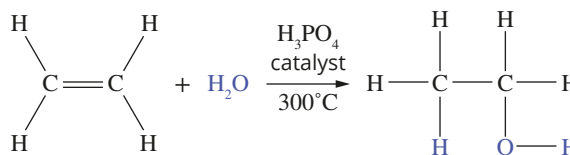


FIGURE 8.4.8 Addition reaction of ethene with water in the presence of a phosphoric acid catalyst

The reaction is carried out at 300°C. The gaseous reactants are passed over a solid bed of the catalyst and gaseous ethanol is formed.

The reaction of ethene with steam is often described as a hydration reaction. Hydration reactions are reactions that involve water as a reactant. In the addition reaction, water is 'added' across the double bond. The reaction is used for the commercial manufacture of ethanol because it is a one-step process that uses little energy, apart from initial heating. The catalyst is a solid, whereas the reactants and products are gases, which means it is easy for manufacturers to remove the product from the reaction mixture, leaving the catalyst intact.

EXTENSION

Markovnikov's rule

Markovnikov's rule describes the way that asymmetrical reagents, such as hydrogen halides (HCl, HBr and HI) add to asymmetrical alkenes. This rule states that the electron-deficient atom (the hydrogen atom due to its lower electronegativity) will add to the carbon atom with more hydrogen atoms already bonded to it, and the electron-rich atom (the halogen atom due to its higher electronegativity) will add to the carbon atom with fewer hydrogen atoms already bonded to it.

For example, when HCl is added to prop-1-ene, there are two possible products. The product that is more likely to be formed involves the hydrogen atom being added to the first carbon atom, and the chlorine atom being added to the second carbon atom, as shown in Figure 8.4.9.

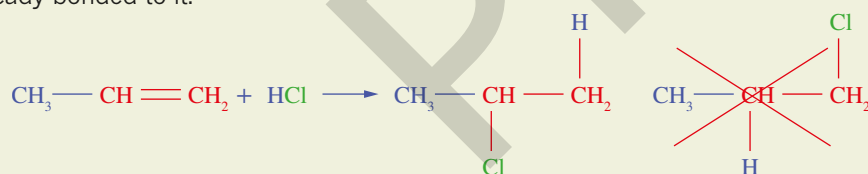


FIGURE 8.4.9 When HCl is added to propene, the H and Cl atoms are added to each side of the double bond. The hydrogen atom is added to the carbon atom with the greater number of hydrogen atoms already attached, according to Markovnikov's rule.

Test for unsaturation

Bromine is used in a standard test to determine whether a substance contains molecules that are unsaturated (contain carbon-carbon double bonds). An aqueous bromine solution is orange in colour. When a few drops are added to an alkane, which is a saturated compound, there is no immediate reaction and the colour does not change. This is because substitution reactions are slow and require a catalyst. However, when a few drops are added to an unsaturated compound, such as an alkene, an addition reaction occurs and the orange colour disappears almost instantly as the bromine is consumed (Figure 8.4.10).



FIGURE 8.4.10 Test for unsaturation. Adding a few drops of orange-coloured bromine solution to hexane (right) produces no immediate reaction. In sunflower oil (left), the colour of the bromine solution disappears almost immediately because molecules in the sunflower oil contain carbon-carbon double bonds, which undergo addition reactions with bromine (Br₂).

CHEMFILE

Addition polymers

Under some conditions, alkenes undergo an addition reaction with themselves to produce long chains, known as addition polymers. Polymers are long molecules that contain hundreds or thousands of smaller repeating units called monomers. The reaction of the monomer ethene with itself to form polyethene, shown in Figure 8.4.11, is the simplest example of the addition polymerisation process. Several thousand ethene monomers usually react to make one molecule of polyethene.

Large square brackets, and the subscript n , are used to simplify the drawing of long polymer molecules. The value of n may vary within each polymer molecule, but the average molecular chain formed might contain as many as 20 000 carbon atoms. Polymers really are very large molecules!

Polypropene, polyvinyl chloride (PVC) and polystyrene are all addition polymers that are commonly found in a range of different applications (Figure 8.4.12).

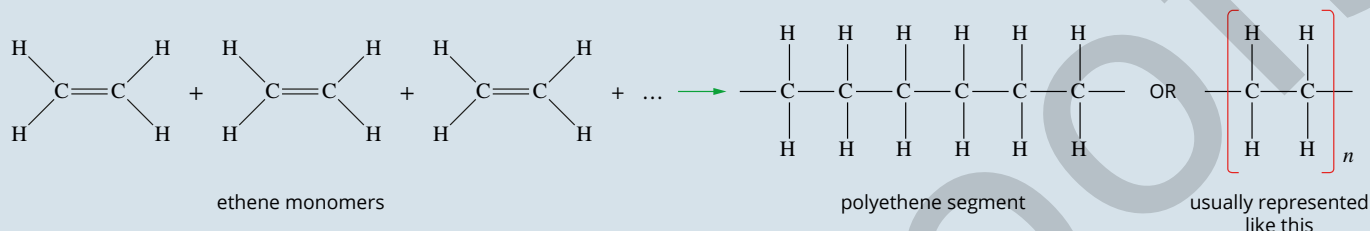


FIGURE 8.4.11 Thousands of ethene monomers join together to make one chain of polyethene.

a	propene	polypropene	
			Durable, cheap Artificial grass for indoor courts, dishwasher-safe plastic, ice-cream containers, rope
b	dichloroethene	polyvinylidene chloride (PVDC)	
			Food wrap
c	phenylethene (styrene)	polyphenylethene (polystyrene)	
			Hard, brittle, good at low temperatures Toys, expanded foam
d	methyl cyanoacrylate	polymethylcyanoacrylate (Superglue)	
			Polymerised by water Superglue

FIGURE 8.4.12 Applications of common addition polymers. (a) polypropene, (b) polyvinylidene chloride (PVDC), (c) polystyrene, (d) polymethylcyanoacrylate

CHEMISTRY IN ACTION

Making margarine

The presence of double bonds (unsaturation) in long hydrocarbon molecules decreases their melting and boiling points. Vegetable oils such as olive oil and canola oil contain high proportions of polyunsaturated and mono-unsaturated fats and as a result are liquids at room temperature.

To prepare margarine, the melting point of vegetable oils can be increased, thereby hardening them, through using an addition reaction. By reacting the oil with hydrogen gas in the presence of a nickel catalyst, the double bonds are broken and replaced with single carbon-carbon bonds. The reaction conditions are carefully controlled so that some, not all, of the carbon-carbon double bonds are broken. In this way, the consistency and texture of the margarine at room temperature can be controlled (Figure 8.4.13).



FIGURE 8.4.13 Vegetable oils contain a high degree of unsaturation (many double bonds) and therefore have low melting points and are liquids at room temperature. By using an addition reaction with hydrogen, the double bonds can be broken, increasing the melting point and creating a product with the texture and consistency of margarine.

COMBUSTION

Hydrocarbons are used widely as fuels. The reaction between a fuel and oxygen is a combustion reaction. The burning of petrol in a car engine and the use of natural gas for cooking are examples of combustion.

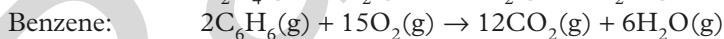
Complete combustion

If the supply of oxygen is plentiful, the products of combustion will be carbon dioxide and water. This is called **complete combustion**. Combustion reactions release significant amounts of energy, which is why they are popular as fuels. Their use as fuels is discussed further in Chapter 11.

The equations for the complete combustion of two alkanes, methane and propane are:



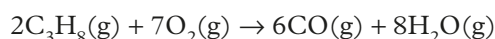
Alkenes and benzene can also undergo combustion reactions. The equations for the combustion of ethene and benzene, for example, are:



Incomplete combustion

When the supply of oxygen is limited, carbon monoxide, which is poisonous, can be formed instead of carbon dioxide. This can be dangerous when running engines in confined spaces. Carbon monoxide is hard to detect because it is colourless and odourless. Workers cleaning out industrial tanks, for example, can sometimes collapse, not realising that a dangerous level of carbon monoxide is present.

The following equation shows the **incomplete combustion** of propane:



8.4 Review

SUMMARY

- Alkanes undergo substitution reactions with halogens in the presence of UV light.
- Benzene undergoes substitution reactions with halogens and alkyl halides in the presence of a catalyst.
- Alkenes undergo addition reactions with halogens, hydrogen, hydrogen halides and water. The double bond is broken resulting in the formation of a single C–C bond.
- Alkanes and alkenes undergo combustion reactions. In excess oxygen, these combustion reactions produce carbon dioxide and water vapour, whereas in limited oxygen, carbon monoxide is produced instead of carbon dioxide.

KEY QUESTIONS

- Which one of the following describes the reaction that occurs between an alkane and chlorine in the presence of ultraviolet light?
A combustion reaction
B substitution reaction
C elimination reaction
D addition reaction
- Write a balanced equation for a reaction between methane and Br_2 in the presence of UV light.
- Write a balanced equation for the reaction between but-1-ene and Cl_2 .
- Write a balanced equation for the reaction between benzene and Br_2 in the presence of a suitable catalyst.
- If bromine is readily decolourised by a particular compound, which one of the following could be a possible identity of that compound?
A propane
B 4-methyl-pent-2-ene
C methyl-benzene
D 3-methyl-pentane

Chapter review

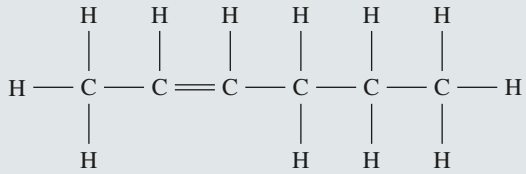
KEY TERMS

addition reaction	crude oil	sediment
alkane	excess reactant	side group
alkene	fossil fuel	stem name
alkyl group	geometric isomers	structural formula
complete combustion	homologous series	structural isomer
condensed structural formula	hydrocarbon	unsaturated
cracking	incomplete combustion	
	saturated	

Alkanes

- Why can carbon form so many compounds?
- Which one of the following best describes a homologous series?
 - compounds in which successive members differ by one carbon and two hydrogen atoms
 - a series of compounds that exist in different physical forms
 - compounds in which each member differs from the previous one by a $-\text{CH}_3$ group
 - compounds with the same molecular formula but different arrangements of atoms
- The formula of a hydrocarbon is $\text{C}_{16}\text{H}_{34}$.
 - To which homologous series does it belong?
 - What is the formula of the next hydrocarbon in the homologous series?
 - What is the formula of the previous hydrocarbon in the same homologous series?
 - What mass of carbon is present in 275 g of the hydrocarbon $\text{C}_{16}\text{H}_{34}$?
- Draw the structural formulae and give the systematic names for each of the following.
 - CH_3CH_3
 - $\text{CH}_3\text{CH}(\text{CH}_3)_2$
 - $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$
 - $\text{CH}_3(\text{CH}_2)_3\text{CH}_3$
- Draw the structural formulae of the following alkanes.
 - 3-methylpentane
 - 2,2-dimethylbutane
 - 2,3-dimethylpentane
- Hexane is a liquid at room temperature but propane is a gas. Explain why.

Alkenes

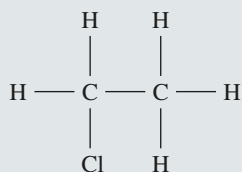
- Classify each of the following hydrocarbons as alkanes or alkenes.
 - C_2H_6
 - C_3H_6
 - $\text{C}_{20}\text{H}_{42}$
 - C_5H_{10}
 - C_5H_{12}
- Draw the structural formulae and give the systematic names of all the isomers of butene.
- Write the systematic names of the following unsaturated hydrocarbons.
 - 
 - $\text{CH}_3\text{CHCHCH}(\text{CH}_3)\text{CH}_3$
 - C_3H_6

Benzene

- Draw the Kekulé structure of benzene and describe why it is an inaccurate representation.
- Which of the following is most likely to contain an aromatic (benzene) ring?
 - $\text{C}_6\text{H}_{11}\text{Br}$
 - $\text{C}_6\text{H}_5\text{Br}$
 - C_6H_{14}
 - $\text{C}_6\text{H}_6\text{Br}_6$

Reactions of hydrocarbons

- 12** Write a balanced chemical equation for the complete combustion of the following alkanes. Remember to include the states of the reactants and products.
- gaseous methane
 - liquid hexane
 - gaseous methylpropane
 - solid $C_{31}H_{64}$
- 13** Write a balanced chemical equation for each of the following reactions.
- complete combustion of ethene gas
 - gaseous propene bubbled into aqueous bromine solution
 - gaseous but-2-ene reacted with steam in the presence of a phosphoric acid catalyst
 - gaseous ethene reacted with gaseous hydrogen chloride.
- 14** You are given two hydrocarbons, hexane and hex-2-ene. Describe a chemical test by which you could distinguish between them.
- 15** The compound below can be synthesised (made) by using either an addition or a substitution reaction. Write equations to represent both methods.



- 16** Butane is reacted to completion (all the hydrogen atoms are substituted with Br atoms) with an excess of bromine water in the presence of UV light. How many molecules of bromine are needed for each molecule of butane used?
- 17** Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) can be produced from ethene using an addition reaction. Write an equation to represent this reaction.

Connecting the main ideas

- 18** Explain the following.
- The first member of the alkene homologous series is ethene, not methene.
 - Carbon compounds usually have four covalent bonds around each carbon atom.
- 19** Labels on margarine and oil often include one of the following terms: polyunsaturated, mono-unsaturated, saturated. What does each term mean?
- 20** A compound is found to have the formula $\text{C}_{10}\text{H}_{20}$. What would you expect to observe if a few drops of bromine water were added?